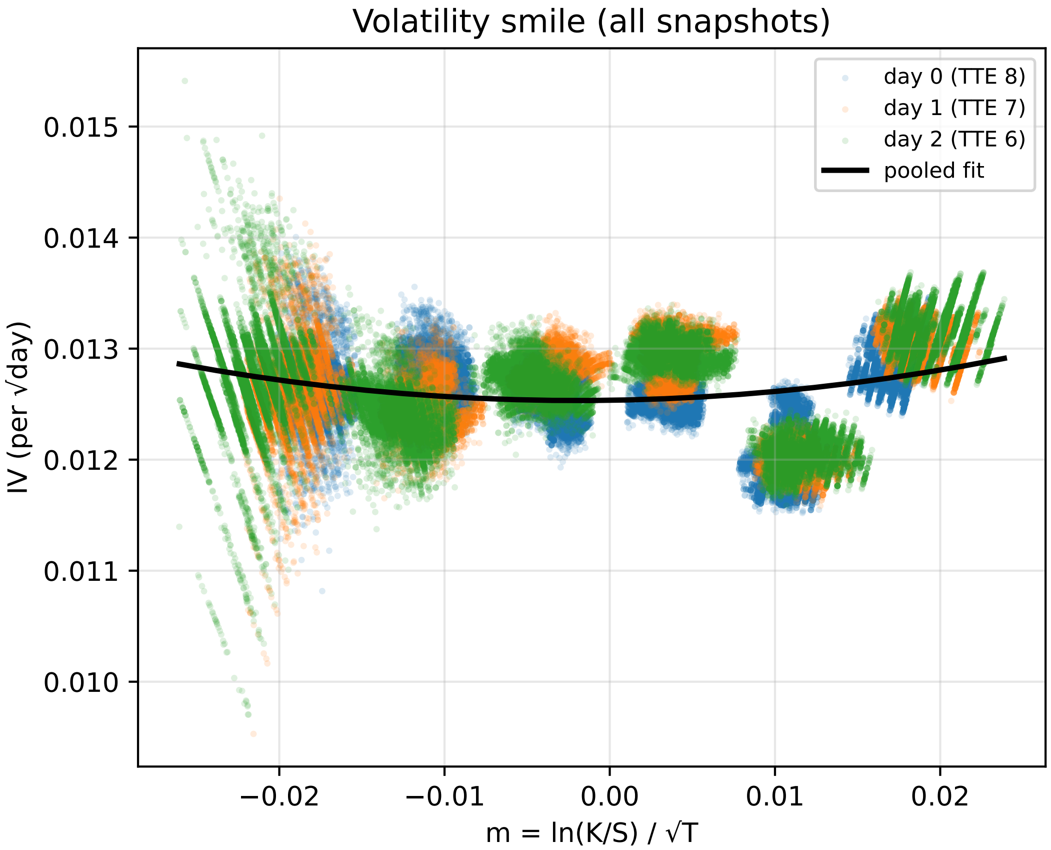
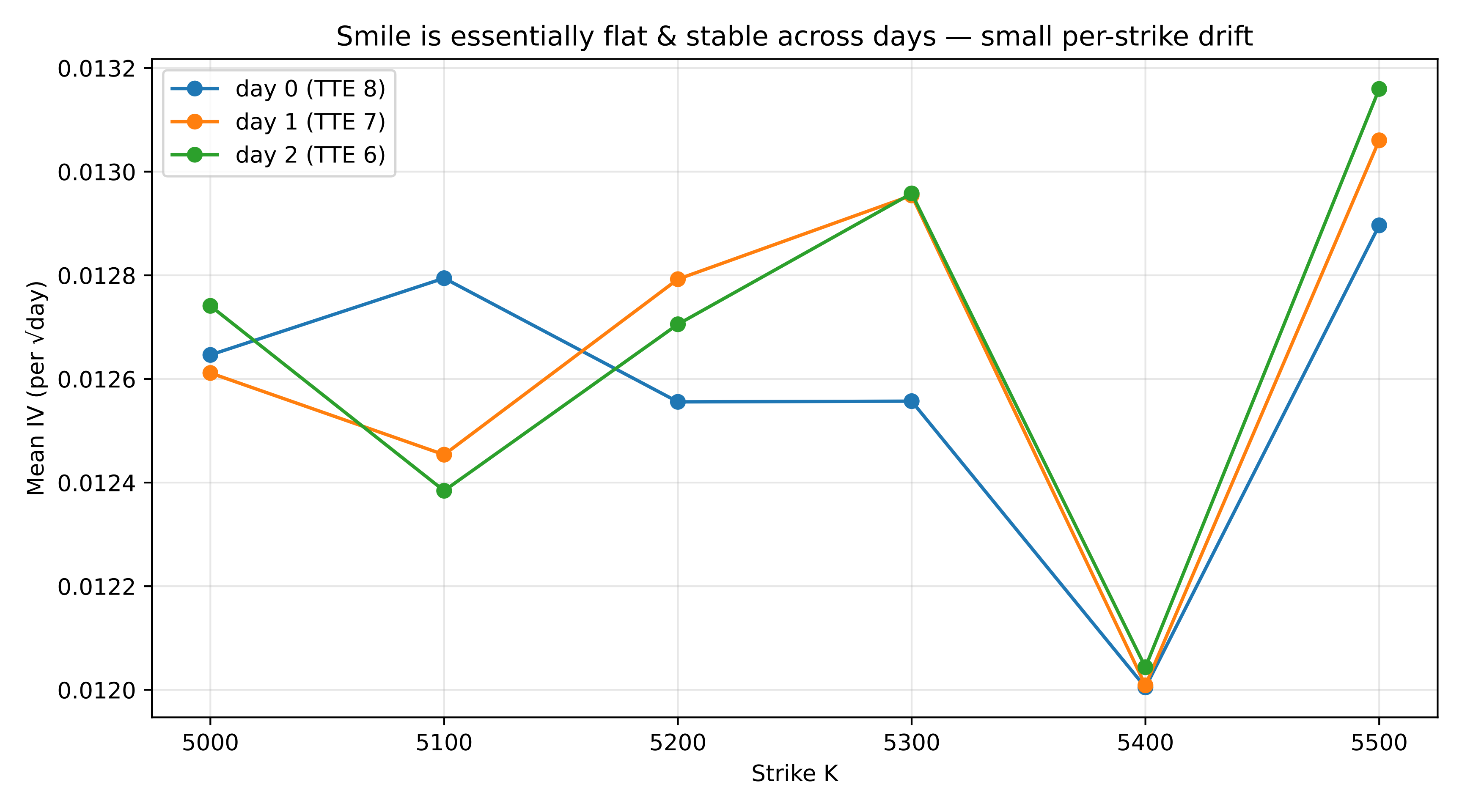
Round 3 — IV smile across live strikes (days 0/1/2)

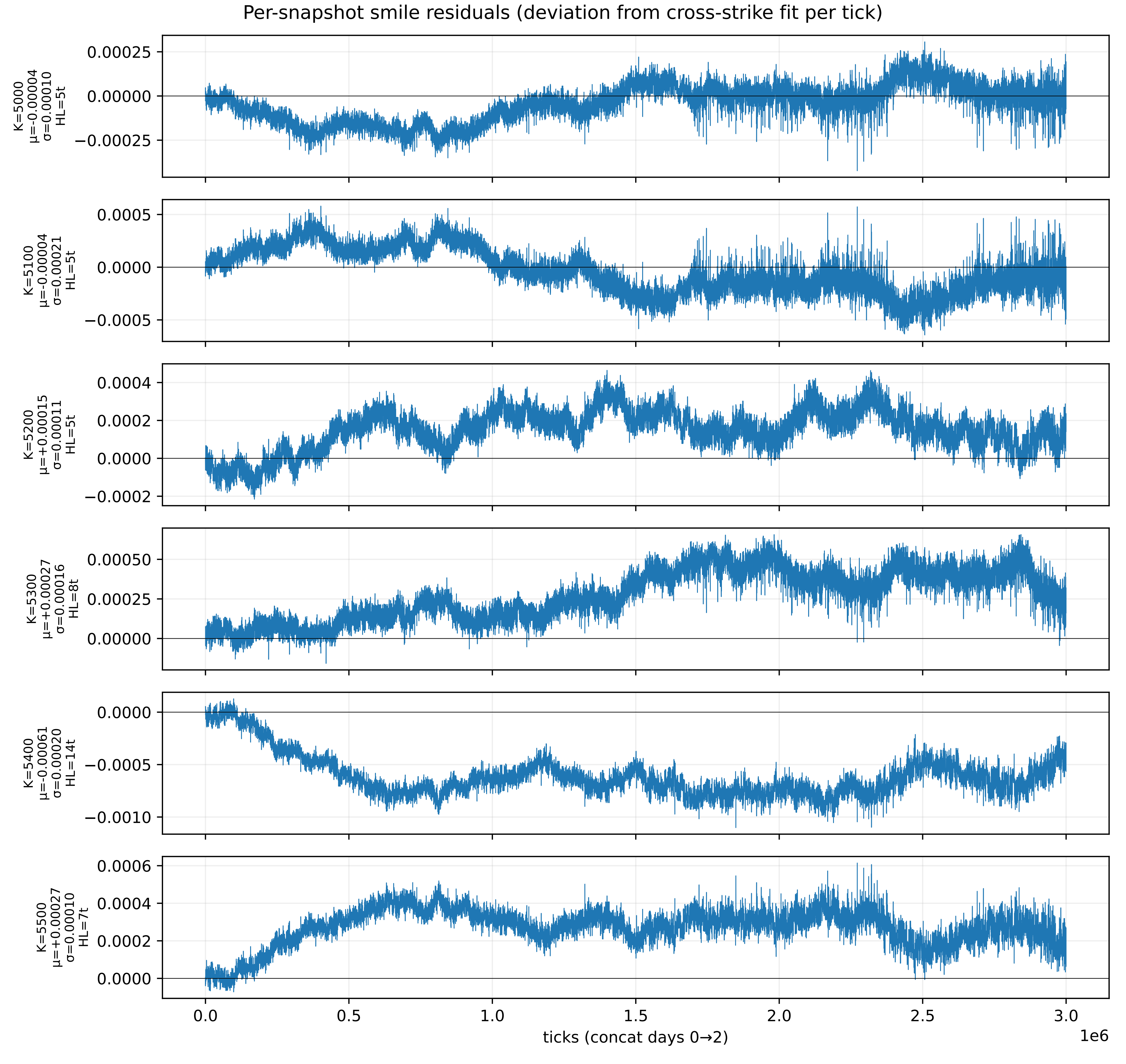


Per-strike IV summary

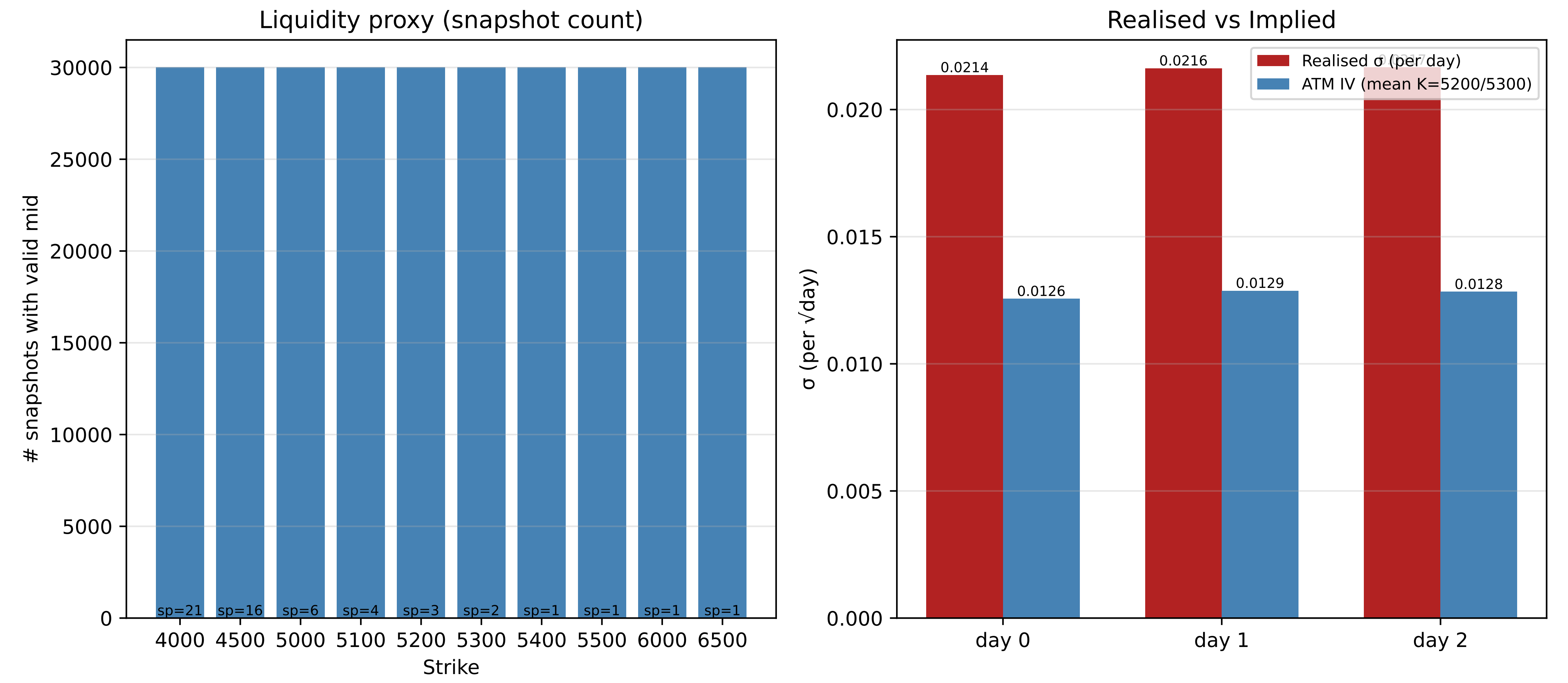
K	mean	std	count	resid_mean	resid_snap_mean
5000	0.012666	0.000374	29976	-0.000039	-0.000045
5100	0.012544	0.000266	30000	-0.000041	-0.000042
5200	0.012685	0.000180	30000	0.000149	0.000151
5300	0.012823	0.000231	30000	0.000271	0.000274
5400	0.012019	0.000194	30000	-0.000612	-0.000609
5500	0.013039	0.000219	30000	0.000270	0.000271

Quad fit: $IV(m) = 0.56799 \cdot m^2 + 0.00225 \cdot m + 0.01253$


Smile is essentially flat & stable across days — small per-strike drift
day 0 (TTE 8)
day 1 (TTE 7)
day 2 (TTE 6)
Mean IV (per √day)
0.0132
0.0130
0.0128
0.0126
0.0124
0.0122
0.0120
5000
5100
5200
5300
5400
5500
Strike K



Per-snapshot smile residuals (deviation from cross-strike fit per tick)
K=5000
μ=-0.00004
σ=0.00010
HL=5t
K=5100
μ=-0.00004
σ=0.00021
HL=5t
K=5200
μ=+0.00015
σ=0.00011
HL=5t
K=5300
μ=+0.00027
σ=0.00016
HL=8t
K=5400
μ=-0.00061
σ=0.00020
HL=14t
K=5500
μ=+0.00027
σ=0.00010
HL=7t
ticks (concat days 0→2)
1e6



Liquidity proxy (snapshot count)
snapshots with valid mid
0
5000
10000
15000
20000
25000
30000
sp=21
sp=16
sp=6
sp=4
sp=3
sp=2
sp=1
sp=1
sp=1
sp=1
4000
4500
5000
5100
5200
5300
5400
5500
6000
6500
Strike
Realised vs Implied
σ (per √day)
0.000
0.005
0.010
0.015
0.020
Realised σ (per day)
ATM IV (mean K=5200/5300)
0.0214
0.0126
0.0216
0.0129
0.0128
day 0
day 1
day 2

SIGNAL vs NOISE — per-strike residual diagnostics

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    K   mean_res     sd_res   t-stat   AR(1)   HL(t)   edge_PnL  verdict
------------------------------------------------------------------------------
 5000  -0.000045   0.000104    -74.2   0.861       5        0.3  MEAN-REVERT
 5100  -0.000042   0.000213    -34.1   0.881       5        1.5  MEAN-REVERT
 5200  +0.000151   0.000108   +241.3   0.877       5        1.1  MEAN-REVERT
 5300  +0.000274   0.000160   +296.2   0.916       8        1.7  MEAN-REVERT
 5400  -0.000609   0.000200   -527.7   0.952      14        1.5  MEAN-REVERT
 5500  +0.000271   0.000097   +482.5   0.905       7        0.4  MEAN-REVERT

Interpretation:
  • t-stat > 3 ⇒ residual mean is statistically non-zero (real drift)
  • AR(1) > 0 ⇒ residual is autocorrelated (slow-moving, tradeable)
  • HL = half-life in ticks (lower = faster reversion = more cycles/day)
  • edge_PnL = 2·σ_resid·vega — ballpark seashells per round-trip if you fade
    the residual at +1σ and exit at the smile.

Realised σ per day:  {0: 0.021356536275101985, 1: 0.02162128784911193, 2: 0.021655688120964463}
Implied σ per day:   {0: 0.012556421458831662, 1: 0.012873292484511958, 2: 0.012831731772172842}
  → IV under-prices realised; gap matters for the level (vega-PnL on long/short vol).
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